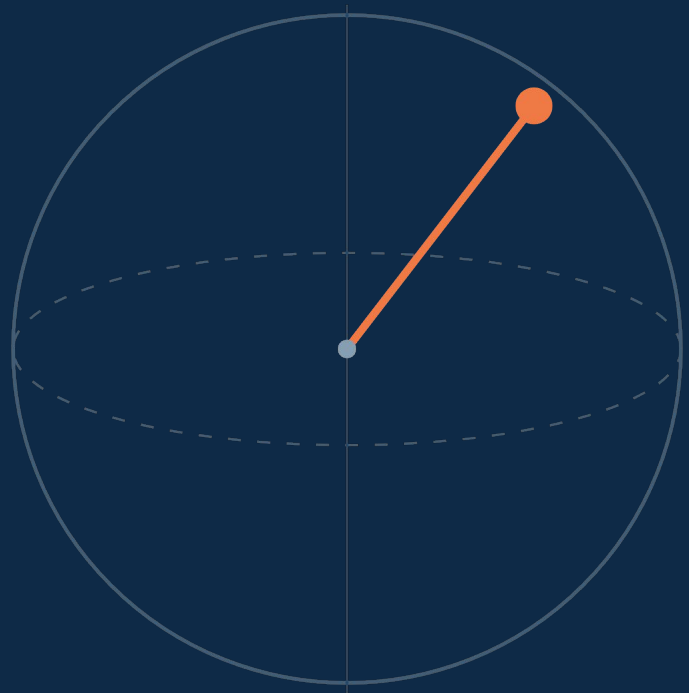
BREAQ HACKATHON

Optimizations on a Quantum ML model

A 4-qubit fraud detector that holds up under noise, at 17 % of the baseline's quantum cost



Cavescu Dumitru-Andrei

THE CHALLENGE

Rank rare fraud on noisy, costly qubits

0.17 %
of 284,807 card payments are fraud, so we rank by AUPRC, not accuracy

p = 0.05
depolarizing noise after every gate, the hardest level tested

1,024
shots per circuit run on a real device; every run costs QPU time

TASK 1 BASELINE

collapses on real hardware

0.58
AUPRC, no noise

0.41
AUPRC at p = 0.05

0.02
AUPRC at p = 0.05 with 1,024 shots

83 h
QPU time to train: 2.9 M runs

Full split, 5 seeds

IDEA 1 OF 5 · COST

“Dead” gates

13 rotations and 3 CNOTs can never reach the measured qubit, and checking validation after every epoch cost more than training itself.

BEFORE

736k

circuit runs: every gate kept, validation every epoch

AFTER

274k

dead gates removed, validation every 5 epochs on half the set

EFFECT

–63 % runs

Same AUPRC: the removed gates could not change the result, not even under noise

Development sample (30 % of the data), 5 seeds

IDEA 2 OF 5 · ENCODING

One rotation per qubit

The ZZ feature map sent each feature round a full circle, so very low and very high values looked alike. RY(x) maps low to high, pole to pole.

AUPRC AT P = 0.05

0.57

ZZ feature map, 30 CNOTs



AUPRC AT P = 0.05

0.75

angle embedding, 6 CNOTs

EFFECT

Biggest AUPRC growth

Noise no longer reshuffles the ranking: Spearman ρ 0.95 at p = 0.05

Development sample (30 % of the data), 5 seeds

IDEA 3 OF 5 · READOUT

Point the circuit at the fraud signal

Rotate the inputs so the measured qubit carries the most fraud-revealing direction, and learn the score's scale with sigmoid(a·⟨Z⟩ + b).

RANKING KEPT, P = 0.05

0.95

Spearman ρ, angle embedding alone



RANKING KEPT, P = 0.05

0.98

fraud axis on qubit 0 and a trainable readout

EFFECT

Maximizes Spearman

a and b are classical numbers: no extra circuit runs.

Development sample (30 % of the data), 5 seeds

IDEA 4 OF 5 · TRAINING

Start near zero

Initial angles in ±0.1 instead of random: the circuit starts by reading the fraud axis, so training begins from the classical ranking.

Random start

0.050

how much AUPRC varies between seeds



Start near zero

0.022

half the variation: the 5 seeds now agree

EFFECT

F1 0.74 → 0.80

at p = 0.05; better on 5 of 5 seeds on untouched data

Development sample (30 % of the data), 5 seeds

IDEA 5 OF 5 · FROM THE LITERATURE

A tailored ansatz

After Dao (2025): keep only the rotations that can change the measured qubit, $\exp(-i\theta Y)$ and $\exp(-i\theta Y_o Z_j)$, in a single layer.

TRAINABLE ANGLES

19

pruned EfficientSU2-style ansatz



TRAINABLE ANGLES

7

tailored evolved-operator ansatz

EFFECT

−46 % runs

Same AUPRC within noise: validation +0.01, untouched data −0.01

Development sample (30 % of the data), 5 seeds · Dao, arXiv:2503.13368

RESULT · FULL COMMON SPLIT, 5 SEEDS

Our VQC vs the Task 1 baseline

0.68

AUPRC at p = 0.05

baseline 0.41

0.64

AUPRC at p = 0.05 with 1,024 shots

baseline 0.02

−83 %

circuit runs to train and test

2.9 M → 0.49 M runs · 83 → 14 QPU hours

5

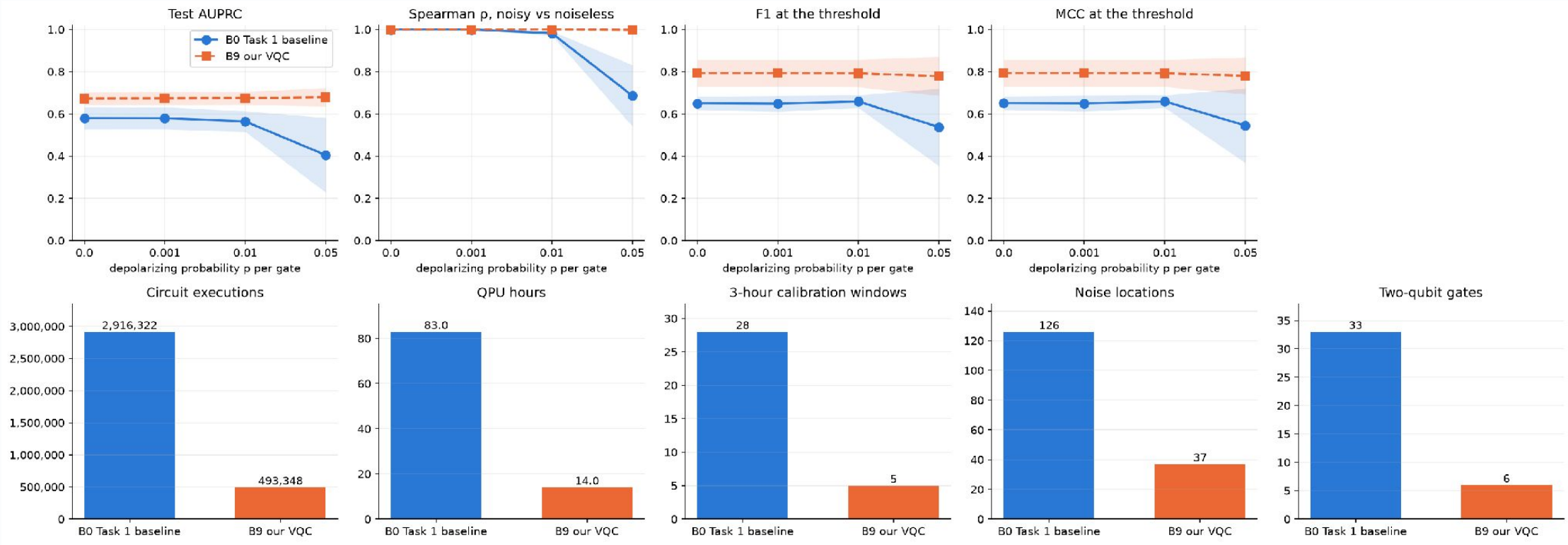
3-hour calibration windows to train

baseline 28: far fewer recalibrations mid-training

Test set: 56,962 payments, 98 frauds · 7 trainable angles and 6 CNOTs, against 32 and 33

RESULT · UNDER NOISE

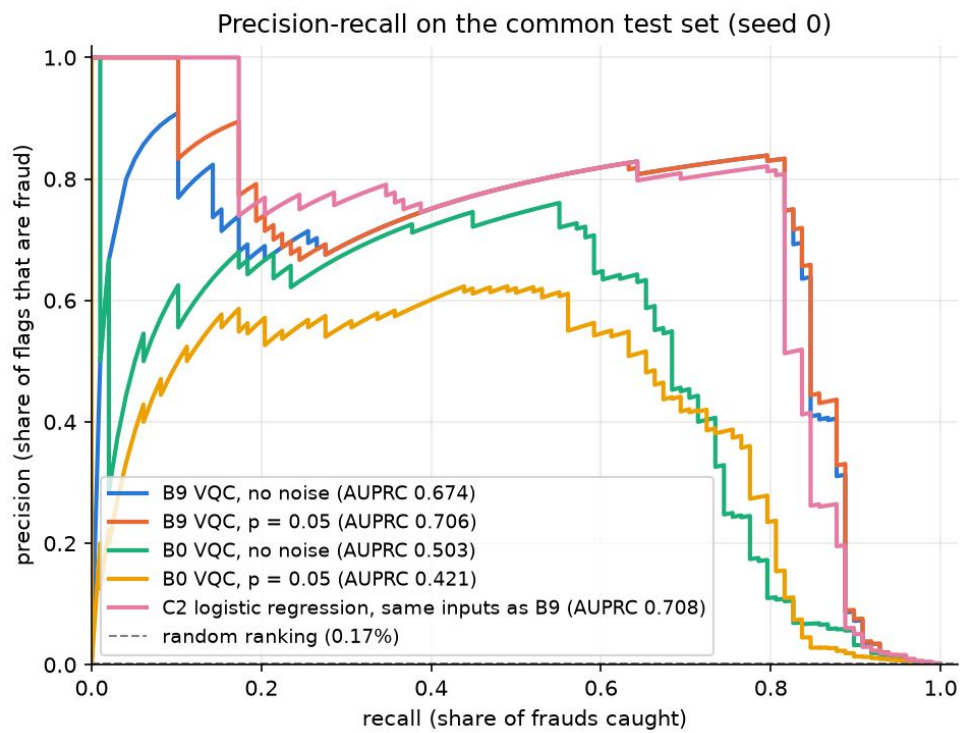
Flat under noise, at a sixth of the cost



Orange: our VQC (B9) · Blue: Task 1 baseline (B0) · full common split, mean ± std over 5 seeds

RESULT · RANKING

Frauds stay on top, even with real shots



Noise barely moves our curve

B9 without noise (blue) and at p = 0.05 (orange) nearly coincide; the baseline (green, yellow) sits far lower.

Real measurement

With 1,024 shots at p = 0.05, over 5 seeds: B9 0.64, baseline 0.02.

The honest bar

Logistic regression on the same inputs (pink) reaches 0.71. B9 is about 0.03 behind, but holds under noise and shots.

Precision-recall on the common test set, seed 0

TAKEAWAYS

What made the difference

01

Encoding first

One rotation per qubit beat every ZZ variant: the biggest single gain.

02

Design for the readout

Gates and angles that can’t reach the measured qubit are pure cost.

03

Check it honestly

Confirmed on 199k untouched payments and with real shot counts. Logistic regression is still 0.03 ahead.

Thank you. Questions?